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Lake Erie - Lake Guardian CSMI Cruises, All results are in ug/L

ESF ID #	Sample ID	Collection Date	Sample Vol. (ml)	Total Microcystins	MDL	Total Anatoxin	MDL	Cylindro MDL
24-1861	1 - AAM_ERI_20240814_15_PT_ET_1_GFF	8/14/2024	2340		(<0.01)		(<0.010)	(<0.000)
24-1862	2 - AAM_ERI_20240813_28_PT_ET_1_GFF	8/13/2024	2755	0.01	(<0.01)		(<0.008)	(<0.000)
24-1863	3 - AAM_ERI_20240812_38_PT_ET_1_GFF	8/12/2024	2520		(<0.01)		(<0.009)	(<0.000)
24-1864	4 - AAM_ERI_20240812_43_PT_ET_1_GFF	8/12/2024	2460		(<0.01)		(<0.009)	(<0.000)
24-1865	5 - AAM_ERI_20240810_59_PT_ET_1_GFF	8/10/2024	1940	0.29	(<0.00)		(<0.006)	(<0.002)
24-1866	6 - AAM_ERI_20240809_60_PT_ET_1_GFF	8/9/2024	1550		(<0.01)		(<0.008)	(<0.003)
24-1867	7 - AAM_ERI_20240809_61_PT_ET_1_GFF	8/9/2024	2680	0.08	(<0.00)		(<0.005)	(<0.002)
24-1868	8 - AAM_ERI_20240812_73_PT_ET_1_GFF	8/12/2024	2660		(<0.00)		(<0.005)	(<0.002)
24-1869	9 - AAM_ERI_20240813_78_PT_ET_1_GFF	8/13/2024	2600		(<0.00)		(<0.005)	(<0.002)
24-1870	10 - AAM_ERI_20240810_91_PT_ET_1_GFF	8/10/2024	2650	0.07	(<0.00)		(<0.005)	(<0.002)
24-1871	11 - AAM_ERI_20240813_95_PT_ET_1_GFF	8/13/2024	2630		(<0.00)		(<0.005)	(<0.002)
24-1872	12 - AAM_ERI_20240814_932_PT_ET_1_GFF	8/14/2024	2490		(<0.00)		(<0.005)	(<0.002)
24-1873	13 - AAM_ERI_20240814_936_PT_ET_1_GFF	8/14/2024	2440		(<0.00)		(<0.005)	(<0.002)
24-1874	14 - AAM_ERI_20240814_937_PT_ET_1_GFF	8/14/2024	2650		(<0.00)		(<0.005)	(<0.002)
24-1875	15 - AAM_ERI_20240812_951_PT_ET_1_GFF	8/12/2024	2715		(<0.00)	0.003	(<0.005)	(<0.002)
24-1876	16 - AAM_ERI_20240810_958_PT_ET_1_GFF	8/10/2024	2740		(<0.00)		(<0.005)	(<0.002)
24-1877	17 - AAM_ERI_20240810_964_PT_ET_1_GFF	8/10/2024	2860	0.03	(<0.00)	0.002	(<0.004)	(<0.002)
24-1878	18 - AAM_ERI_20240810_966_PT_ET_1_GFF	8/10/2024	2710	0.14	(<0.00)		(<0.005)	(<0.002)
24-1879	19 - AAM_ERI/ONT_20240815_PT_ET_1_GFF	8/15/2024	2190		(<0.00)		(<0.006)	(<0.002)
24-1880	20 - AAM_ERI_20240916_09_PT_ET_1_GFF	9/16/2024	2540		(<0.00)		(<0.005)	(<0.002)
24-1881	21 - AAM_ERI_20240914_G09_PT_ET_1_GFF	9/14/2024	2920	0.07	(<0.00)	0.048	(<0.004)	(<0.002)
24-1882	22 - AAM_ERI_20240916_10_PT_ET_1_GFF	9/16/2024	2340		(<0.00)	0.001	(<0.005)	(<0.002)
24-1883	23 - AAM_ERI_20240915_15_PT_ET_1_GFF	9/15/2024	2340		(<0.00)		(<0.005)	(<0.002)
24-1884	24 - AAM_ERI_20240915_22_PT_ET_1_GFF	9/15/2024	2700		(<0.00)		(<0.005)	(<0.002)
24-1885	25 - AAM_ERI_20240915_28_PT_ET_1_GFF	9/15/2024	2580		(<0.00)		(<0.005)	(<0.002)

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Lake Erie - Lake Guardian CSMI Cruises, page 2, All results are in ug/L

ESF ID #	Sample ID	Collection Date	Sample Vol. (ml)	Total Microcystins	MDL	Total Anatoxin	MDL	Cylindro MDL
24-1886	26 - AAM_ERI_20240914_38_PT_ET_1_GFF	9/14/2024	2580		(<0.00)		(<0.005)	(<0.002)
24-1887	27 - AAM_ERI_20240914_43_PT_ET_1_GFF	9/14/2024	2600		(<0.00)		(<0.005)	(<0.002)
24-1888	28 - AAM_ERI_20240913_58_PT_ET_1_GFF	9/13/2024	2800	0.13	(<0.00)		(<0.004)	(<0.002)
24-1889	29 - AAM_ERI_20240913_59_PT_ET_1_GFF	9/13/2024	2700	0.07	(<0.00)		(<0.005)	(<0.002)
24-1890	30 - AAM_ERI_20240913_60_PT_ET_1_GFF	9/13/2024	2600	0.03	(<0.00)		(<0.005)	(<0.002)
24-1891	31 - AAM_ERI_20240913_61_PT_ET_1_GFF	9/13/2024	2600	0.02	(<0.00)		(<0.005)	(<0.002)
24-1892	32 - AAM_ERI_20240914_73_PT_ET_1_GFF	9/14/2024	3000		(<0.00)		(<0.004)	(<0.002)
24-1893	33 - AAM_ERI_20240915_78_PT_ET_1_GFF	9/15/2024	2740		(<0.00)		(<0.005)	(<0.002)
24-1894	34 - AAM_ERI_20240914_91_PT_ET_1_GFF	9/14/2024	2760	0.02	(<0.00)		(<0.005)	(<0.002)
24-1895	35 - AAM_ERI_20240915_95_PT_ET_1_GFF	9/15/2024	2700		(<0.01)		(<0.005)	(<0.002)
24-1896	36 - AAM_ERI_20240916_932_PT_ET_1_GFF	9/16/2024	2500		(<0.01)		(<0.005)	(<0.002)
24-1897	37 - AAM_ERI_20240916_936_PT_ET_1_GFF	9/16/2024	2320		(<0.01)		(<0.005)	(<0.002)
24-1898	38 - AAM_ERI_20240915_937_PT_ET_1_GFF	9/15/2024	2040		(<0.01)		(<0.006)	(<0.002)
24-1899	39 - AAM_ERI_20240914_951_PT_ET_1_GFF	9/14/2024	2600	0.01	(<0.01)		(<0.005)	(<0.002)
24-1900	40 - AAM_ERI_20240913_958_PT_ET_1_GFF	9/13/2024	2600	0.02	(<0.01)	0.003	(<0.005)	(<0.002)
24-1901	41 - AAM_ERI_20240914_959_PT_ET_1_GFF	9/14/2024	2840	0.01	(<0.01)	0.007	(<0.004)	(<0.002)
24-1902	42 - AAM_ERI_20240913_964_PT_ET_1_GFF	9/13/2024	2700	0.01	(<0.01)		(<0.005)	(<0.002)
24-1903	43 - AAM_ERI_20240914_966_PT_ET_1_GFF	9/14/2024	2695	0.03	(<0.01)	0.004	(<0.005)	(<0.002)
24-1904	44 - AAM_ERI_20240915_EB06_PT_ET_1_GFF	9/15/2024	2585	0.00	(<0.01)		(<0.005)	(<0.002)
24-1905	45 - AAM_ERI_20240916_000_PT_NT_1_GFF	9/16/2024	1460		(<0.02)		(<0.009)	(<0.003)
24-1906	46 - AAM_ERI_20240914_91_PT_ET_1_GFF_Sed	9/14/2024	2000	0.02	(<0.01)		(<0.006)	(<0.002)
24-1907	47 - AAM_ERI_20240814_15_PT_ET_1_PC	8/14/2024	385		(<0.05)		(<0.059)	(<0.003)
24-1908	48 - AAM_ERI_20240813_28_PT_ET_1_PC	8/13/2024	420		(<0.05)		(<0.054)	(<0.002)
24-1909	49 - AAM_ERI_20240812_38_PT_ET_1_PC	8/12/2024	380		(<0.06)		(<0.060)	(<0.003)
24-1910	50 - AAM_ERI_20240812_43_PT_ET_1_PC	8/12/2024	360		(<0.06)		(<0.063)	(<0.003)

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Lake Erie - Lake Guardian CSMI Cruises, page 3, All results are in ug/L

ESF ID #	Sample ID	Collection Date	Sample Vol. (ml)	Total Microcystins	MDL	Total Anatoxin	MDL	Cylindro MDL
24-1911	51 - AAM_ERI_20240810_59_PT_ET_1_PC	8/10/2024	151	0.12	(<0.14)		(<0.151)	(<0.007)
24-1912	52 - AAM_ERI_20240809_60_PT_ET_1_PC	8/9/2024	470		(<0.04)		(<0.049)	(<0.002)
24-1913	53 - AAM_ERI_20240809_61_PT_ET_1_PC	8/9/2024	295		(<0.07)		(<0.077)	(<0.003)
24-1914	54 - AAM_ERI_20240812_73_PT_ET_1_PC	8/12/2024	110		(<0.19)		(<0.207)	(<0.009)
24-1915	55 - AAM_ERI_20240813_78_PT_ET_1_PC	8/13/2024	348		(<0.06)		(<0.066)	(<0.003)
24-1916	56 - AAM_ERI_20240810_91_PT_ET_1_PC	8/10/2024	310		(<0.07)		(<0.074)	(<0.003)
24-1917	57 - AAM_ERI_20240813_95_PT_ET_1_PC	8/13/2024	350		(<0.06)		(<0.065)	(<0.003)
24-1918	58 - AAM_ERI_20240814_932_PT_ET_1_PC	8/14/2024	510		(<0.04)		(<0.045)	(<0.002)
24-1919	59 - AAM_ERI_20240814_936_PT_ET_1_PC	8/14/2024	370		(<0.06)		(<0.049)	(<0.002)
24-1920	60 - AAM_ERI_20240814_937_PT_ET_1_PC	8/14/2024	320		(<0.07)		(<0.057)	(<0.002)
24-1921	61 - AAM_ERI_20240812_951_PT_ET_1_PC	8/12/2024	330		(<0.06)		(<0.055)	(<0.002)
24-1922	62 - AAM_ERI_20240810_958_PT_ET_1_PC	8/10/2024	105		(<0.20)		(<0.174)	(<0.007)
24-1923	63 - AAM_ERI_20240810_964_PT_ET_1_PC	8/10/2024	240	0.02	(<0.09)		(<0.076)	(<0.003)
24-1924	64 - AAM_ERI_20240810_966_PT_ET_1_PC	8/10/2024	340	0.03	(<0.06)		(<0.054)	(<0.002)
24-1925	65 - AAM_ERI_20240916_09_PT_ET_1_PC	9/16/2024	475		(<0.05)		(<0.038)	(<0.002)
24-1926	66 - AAM_ERI_20240914_G09_PT_ET_1_PC	9/14/2024	270		(<0.08)		(<0.068)	(<0.003)
24-1927	67 - AAM_ERI_20240916_10_PT_ET_1_PC	9/16/2024	450		(<0.05)		(<0.041)	(<0.002)
24-1928	68 - AAM_ERI_20240915_15_PT_ET_1_PC	9/15/2024	340		(<0.06)		(<0.054)	(<0.002)
24-1929	69 - AAM_ERI_20240915_22_PT_ET_1_PC	9/15/2024	213		(<0.10)		(<0.086)	(<0.004)
24-1930	70 - AAM_ERI_20240915_28_PT_ET_1_PC	9/15/2024	500		(<0.04)		(<0.037)	(<0.002)
24-1931	71 - AAM_ERI_20240914_38_PT_ET_1_PC	9/14/2024	360	0.03	(<0.05)		(<0.035)	(<0.013)
24-1932	72 - AAM_ERI_20240914_43_PT_ET_1_PC	9/14/2024	460	0.03	(<0.04)		(<0.027)	(<0.010)
24-1933	73 - AAM_ERI_20240913_58_PT_ET_1_PC	9/13/2024	180		(<0.10)		(<0.069)	(<0.025)
24-1934	74 - AAM_ERI_20240913_59_PT_ET_1_PC	9/13/2024	165		(<0.11)		(<0.076)	(<0.027)
24-1935	75 - AAM_ERI_20240913_60_PT_ET_1_PC	9/13/2024	300	0.03	(<0.06)		(<0.042)	(<0.015)

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Lake Erie - Lake Guardian CSMI Cruises, page 4. All results are in ug/L

ESF ID #	Sample ID	Collection Date	Sample Vol. (ml)	Total Microcystins	MDL	Total Anatoxin	MDL	Cylindro MDL
24-1936	76 - AAM_ERI_20240913_61_PT_ET_1_PC	9/13/2024	255		(<0.07)		(<0.049)	(<0.018)
24-1937	77 - AAM_ERI_20240914_73_PT_ET_1_PC	9/14/2024	105		(<0.17)		(<0.119)	(<0.043)
24-1938	78 - AAM_ERI_20240915_78_PT_ET_1_PC	9/15/2024	385		(<0.05)		(<0.032)	(<0.012)
24-1939	79 - AAM_ERI_20240914_91_PT_ET_1_PC	9/14/2024	190		(<0.10)		(<0.066)	(<0.024)
24-1940	80 - AAM_ERI_20240915_95_PT_ET_1_PC	9/15/2024	375		(<0.05)		(<0.033)	(<0.012)
24-1941	81 - AAM_ERI_20240916_932_PT_ET_1_PC	9/16/2024	575		(<0.03)		(<0.022)	(<0.008)
24-1942	82 - AAM_ERI_20240916_936_PT_ET_1_PC	9/16/2024	740		(<0.02)		(<0.017)	(<0.006)
24-1943	83 - AAM_ERI_20240915_937_PT_ET_1_PC	9/15/2024	540		(<0.03)		(<0.023)	(<0.008)
24-1944	84 - AAM_ERI_20240914_951_PT_ET_1_PC	9/14/2024	320		(<0.06)		(<0.039)	(<0.014)
24-1945	85 - AAM_ERI_20240913_958_PT_ET_1_PC	9/13/2024	320		(<0.06)		(<0.039)	(<0.014)
24-1946	86 - AAM_ERI_20240914_959_PT_ET_1_PC	9/14/2024	245		(<0.07)		(<0.051)	(<0.018)
24-1947	87 - AAM_ERI_20240913_964_PT_ET_1_PC	9/13/2024	290		(<0.06)		(<0.043)	(<0.016)
24-1948	88 - AAM_ERI_20240914_966_PT_ET_1_PC	9/14/2024	225		(<0.08)		(<0.055)	(<0.020)
24-1949	89 - AAM_ERI_20240915_EB06_PT_ET_1_PC	9/15/2024	280		(<0.06)		(<0.045)	(<0.016)
24-1950	90 - AAM_ERI/ONT_20240815_PT_ET_1_PC	8/15/2024	2070		(<0.01)		(<0.006)	(<0.002)
24-1951	91 - AAM_ERI_20240916_000_PT_NT_1_PC	9/16/2024	2280		(<0.01)		(<0.005)	(<0.002)
24-1952	92 - AAM_ERI_20240914_91_PT_ET_1_PC_Sed	9/14/2024	210		(<0.09)		(<0.059)	(<0.021)

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Notes

None of your samples contained cylindrospermopsins. This is typical for the Great Lakes Region where this toxin is either absent or extremely rare

Microcystins were present in a number of samples at low concentrations. Predominate congener was Microcystin LR with an occasional LA or YR. That is a common pattern. Full congener breakdown is given in the separate spreadsheet.

Anatoxins were also detected in low amounts in occasional samples. This is also common for Lake Erie in the past decade. Only the congener Anatoxin-a was detected (bit unusual)

Methods: Sample filters were extracted using ultrasound in 10 ml of 50% methanol using ultrasound, clarified by centrifugation and the toxins determined by chromatography. This extract procedure has been shown to release >90% of the toxins. (Boyer 2007, 2020)

Total Microcystins were determined using HPLC coupled with mass spectrometry. We currently are looking at 22 different congeners (dRR, RR, mRR, H4YR, hYR, YR, dLR, LR, mLR, meLR, AR, FR, WR, dLA, LA, mLA, LL, LY, LW, LF, NOD, and zLR) our of approximately 320 known variants. Individual detection limits are given in the accompanying report and are calculated for each sample individually based on the extraction volume and instrument sensitivity that day. However realize that the MDL is based on a single congener. Multiple microcystins below the MDL could give a total concentration higher that the reported MDL.

Anatoxin-a (ATX), Dihydro-anatoxin-a, Homo-anatoxin-a and Dihydro-homo-anatoxin-a were determined by LC-MS/MS using a modification of US-EPA method 545. To include two confirmation ions. Detections that lacked confirmation with one or the other confirmation ion were excluded. All six anatoxin variants give very similar responses for their primary ions so concentrations were determined using a response factor based on anatoxin A.

Total cylindrospermopsins (CYL), including cylindrospermopsin, epi-cylindrospermopsin and deoxy-cylinderospermopsin, were determined by HPLC coupled with tandem mass spectroscopy (LC-MS/MS) using both a primary quantification ion and two confirmation ions.